Quantum Amplitude Estimation - Lean

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1 Introduction

2 RotationGates

Herein we find definitions of rotation quantum gates, which have not been introduced in previous repositories.

Definition [2.1]
(Rx)

$$R_x(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$$

Definition [2.2]
(Rz)

$$R_z(\theta) = \begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{pmatrix}$$

Definition [2.3]
(Ry)

$$R_y(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$$

3 BHMTK1Tangent

Herein we find a proof from mathematical analysis that will be used later to prove an important theorem regarding Quantum Amplitude Estimation.

Lemma [3.1]
(bhmtTanOddTail_den_pos)
Let $u \in [0, 1)$ *and* $n \in \mathbb{N}$*. Then,*

$$\frac{1}{(2(n+2)-1)^2 - u^2} \geq 0.$$

Lemma [3.2]
(bhmtTanOddTail_term_bound)
Let $u \in [0, 1)$ *and* $n \in \mathbb{N}$*. Then,*

$$\frac{1}{(2(n+2)-1)^2 - u^2} \leq \frac{1}{2(1+u^2)(n+2)(n+1)}$$

Lemma [3.3]
(bhmtInvNatSuccMulSuccSucc_sum_telescope)
Let $N \in \mathbb{N}$*. Then,*

$$\sum_{n=0}^{N-1} \frac{1}{(n+1)(n+2)} = 1 - \frac{1}{N+1}.$$

Lemma [3.4]
(bhmtTanOddTail_summable)
Let $u \in [0, 1)$*. The following series converges.*

$$\sum_{n=0}^{\infty} \frac{1}{(2(n+2)-1)^2 - u^2}.$$

Lemma [3.5]
(bhmtTanOddTail_tsum_le)
Let $u \in [0, 1)$*. Then, the next inequality holds true if the series on the left-hand side converges.*

$$\sum_{n=0}^{\infty} \frac{1}{(2(n+2)-1)^2 - u^2} \leq \frac{1}{2(1+u^2)}.$$

Lemma [3.6]
(bhmtRealCotTerm_summable)
Let $z \in \mathbb{R} \setminus \mathbb{Z}$. The following series converges.

$$\sum_{n=0}^{\infty} \left(\frac{1}{z - (n+1)} + \frac{1}{z + (n+1)} \right).$$

Lemma [3.7]
(bhmtHalfOneMinus_mem_integerComplement)
Let $u \in [0, 1)$. Then, $\frac{1-u}{2} \notin \mathbb{Z}$.

Lemma [3.8]
(bhmtRealCotTerm_summable_of_bounds)
Let $u \in [0, 1)$. The following series converges.

$$\sum_{n=0}^{\infty} \left(\frac{1}{\frac{1-u}{2} - (n+1)} + \frac{1}{\frac{1-u}{2} + (n+1)} \right).$$

Lemma [3.9]
(bhmtTanGroupedTerm_eq_tail)
Let $u \in [0, 1)$ and $n \in \mathbb{N}$. Then,

$$\frac{1}{\frac{1-u}{2} + (n+1)} + \frac{1}{\frac{1-u}{2} - (n+2)} = \frac{4u}{(2(n+2)-1)^2 - u^2}.$$

Lemma [3.10]
(bhmtTanGroupedTerm_summable)
Let $u \in [0, 1)$. The following series converges.

$$\sum_{n=0}^{\infty} \left(\frac{1}{\frac{1-u}{2} + (n+1)} + \frac{1}{\frac{1-u}{2} - (n+2)} \right).$$

Lemma [3.11]
(bhmtTanGroupedTerm_tsum)
Let $u \in [0, 1)$. The following holds if both series converge.

$$\sum_{n=0}^{\infty} \left(\frac{1}{\frac{1-u}{2} + (n+1)} + \frac{1}{\frac{1-u}{2} - (n+2)} \right) = 4u \sum_{n=0}^{\infty} \frac{1}{(2(n+2)-1)^2 - u^2}.$$

Lemma [3.12]
(bhmtRealCotTerm_finite_regroup)
Let $u \in \mathbb{R}$ and $N \in \mathbb{N}$. Then,

$$\begin{aligned}\sum_{n=0}^{N}\left(\frac{1}{\frac{1-u}{2}-(n+1)}+\frac{1}{\frac{1-u}{2}+(n+1)}\right) =& \frac{1}{\frac{1-u}{2}-1} \\ &+\sum_{n=0}^{N-1}\left(\frac{1}{\frac{1-u}{2}+(n+1)}+\frac{1}{\frac{1-u}{2}-(n+2)}\right) \\ &+\frac{1}{\frac{1-u}{2}+(N+1)}.\end{aligned}$$

Lemma [3.13]
(bhmtRealCotTerm_endpoint_tendsto_zero)
Let $u \in \mathbb{R}$. Then,

$$\lim_{N\to\infty}\frac{1}{\frac{1-u}{2}+(N+1)} = 0$$

Lemma [3.14]
(bhmtRealCotTerm_tsum_regroup)
Let $u \in \mathbb{R}$. Assume that the following two series converge.

$$\sum_{n=0}^{\infty}\left(\frac{1}{\frac{1-u}{2}-(n+1)}+\frac{1}{\frac{1-u}{2}+(n+1)}\right),$$
$$\sum_{n=0}^{\infty}\left(\frac{1}{\frac{1-u}{2}+(n+1)}+\frac{1}{\frac{1-u}{2}-(n+2)}\right).$$

Then,

$$\begin{aligned}\sum_{n=0}^{\infty}\left(\frac{1}{\frac{1-u}{2}-(n+1)}+\frac{1}{\frac{1-u}{2}+(n+1)}\right) =& \frac{1}{\frac{1-u}{2}-1} \\ &+\sum_{n=0}^{\infty}\left(\frac{1}{\frac{1-u}{2}+(n+1)}+\frac{1}{\frac{1-u}{2}-(n+2)}\right).\end{aligned}$$

Lemma [3.15]
(bhmtTan_base_poles)
Let $u \in [0,1)$. Then,

$$\frac{1}{\frac{1-u}{2}} + \frac{1}{\frac{1-u}{2} - 1} = \frac{4u}{1-u^2}.$$

Theorem [3.1]
(bhmtTanPartialFractionIdentity_of_bounds)
Let $u \in [0,1)$. If the series of the right-hand side converge, the following holds.

$$\pi \tan\left(u\frac{\pi}{2}\right) = \frac{4u}{1-u^2} + \sum_{n=0}^{\infty} \frac{4u}{(2(n+2)-1)^2 - u^2}.$$

Lemma [3.16]
(deriv_bhmtK1TwoNearestCore_eq_neg_log_factor)
Let $x \in (0, \frac{1}{2})$. Then,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\sin^2(\pi x)}{x^2} + \frac{\sin^2(\pi x)}{(1-x)^2}\right) =& -2\frac{8(1+(1-2x)^2)\cos^2(\pi\frac{1-2x}{2})}{(1-(1-2x)^2)^2} \\ &\times\left(\frac{2(1-2x)}{1+(1-2x)^2} + \frac{4(1-2x)}{1-(1-2x)^2} - \pi\tan\left((1-2x)\frac{\pi}{2}\right)\right).\end{aligned}$$

Lemma [3.17]
(bhmt_log_derivative_expr_nonneg_of_tan_series)
Let $u \in [0,1)$. Then,

$$\frac{2u}{1+u^2} + \frac{4u}{1-u^2} - \pi\tan\left(u\frac{\pi}{2}\right) \geq 0.$$

Theorem [3.2]
(bhmtK1TwoNearestCore_deriv_nonpos)
For all $x \in (0, \frac{1}{2})$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\sin^2(\pi x)}{x^2} + \frac{\sin^2(\pi x)}{(1-x)^2}\right) \leq 0.$$

Theorem [3.3]
(qpeApproxGeometricProbability_two_nearest_lower_bound_k1_of_core_antitone_left)
Let $N \in \mathbb{N}$, $x \in (0,1)$, $N > 0$. *Then,*

$$\frac{\sin^2(\pi x)}{N^2 \sin^2\left(\pi \frac{x}{N}\right)} + \frac{\sin^2(\pi(x-1))}{N^2 \sin^2\left(\pi \frac{x-1}{N}\right)} \geq \frac{8}{\pi^2}.$$

Theorem [3.4]
(qpeCircularPhaseWindowProbability_lower_bound_k1_of_two_adjacent_geometric)
Let $m \in \mathbb{N}$, $\theta, x \in \mathbb{R}$, *and* $y_0, y_1 \in O_{m,1,\theta}$. *Assume* $x \in (0,1)$, *and* $y_0 \neq y_1$.
Assume also that the following two propositions hold.

$$||a(m,\theta,y_0)||^2 = \frac{\sin^2(2^m \pi \frac{x}{2^m})}{2^{2m} \sin^2(\pi \frac{x}{2^m})},$$
$$||a(m,\theta,y_1)||^2 = \frac{\sin^2(2^m \pi \frac{x-1}{2^m})}{2^{2m} \sin^2(\pi \frac{x-1}{2^m})}.$$

Then,

$$Pr\{y \in O_{m,1,\theta} | Y\} \geq \frac{8}{\pi^2}.$$

Theorem [3.5]
(qpeCircularPhaseWindowProbability_lower_bound_k1)
Let $m \in \mathbb{N}$ *and* $\theta \in [0,1]$. *Then,*

$$Pr\{y \in O_{m,1,\theta} | Y\} \geq \frac{8}{\pi^2}.$$

Theorem [3.6]
(bhmt11CircularWindowBound_proved)
Proposition BHMT11CircularWindowBound *in Quantum Phase Estimation section holds.*

4 QuantumPhaseEstimation

Definition [4.1]
(M)

$$\mathrm{M}(m) = 2^m.$$

Lemma [4.1]
(m_neZero)
Let $m \in \mathbb{N}$*. Then,*

$$\mathrm{M}(m) \neq 0.$$

Definition [4.2]
(zeroIndex)

$$0(m) = 0.$$

Definition [4.3]
(uniformState)

$$|u(m)\rangle = \frac{1}{\sqrt{2^m}} \sum_{k=0}^{2^m-1} |k\rangle\,.$$

Definition [4.4]
(phaseState)

$$|p(m,\theta)\rangle = \frac{1}{\sqrt{2^m}} \sum_{k=0}^{2^m-1} e^{2\pi i\theta k}\,|k\rangle\,.$$

Definition [4.5]
(qftMatrix)

$$Q(m) = \frac{1}{\sqrt{2^m}} \sum_{k=0}^{2^m-1} \sum_{l=0}^{2^m-1} e^{2\pi ikl/2^m}\,|k\rangle\,\langle l|\,.$$

Definition [4.6]
(inverseQFTMatrix)

$$IQ(m) = \frac{1}{\sqrt{2^m}} \sum_{k=0}^{2^m-1} \sum_{l=0}^{2^m-1} e^{-2\pi ikl/2^m} |k\rangle \langle l| \, .$$

Definition [4.7]
(qpeApproxAmplitude)

$$a(m, \theta, y) = \langle y| \, IQ(m) \, |p(m, \theta)\rangle \, .$$

Definition [4.8]
(qpeApproxOutcomeProbability)

$$Pr\{y \text{ is measured from } IQ(m) \, |p(m, \theta)\rangle\} = ||a(m, \theta, y)||^2.$$

Definition [4.9]
(qpePhaseWindow)

$$\left|\theta - \frac{y}{2^m}\right| \leq \frac{k}{2^m}.$$

Theorem [4.1]
(qpeApproxOutcomeProbability_nonneg)

$$Pr\{y \text{ is measured from } IQ(m) \, |p(m, \theta)\rangle\} \geq 0.$$

Definition [4.10]
(unitPhaseDistance)

$$d(\theta, \rho) = \min\{|\theta - \rho|, |\theta - \rho + 1|, |\theta - \rho - 1|\}.$$

Theorem [4.2]
(unitPhaseDistance_nonneg)

$$d(\theta, \rho) \geq 0.$$

Theorem [4.3]
(unitPhaseDistance_le_abs_sub)

$$d(\theta, \rho) \leq |\theta - \rho|.$$

Theorem [4.4]
(unitPhaseDistance_le_sub_one)

$$d(\theta, \rho) \leq |\theta - \rho - 1|.$$

Theorem [4.5]
(unitPhaseDistance_le_add_one)

$$d(\theta, \rho) \leq |\theta - \rho + 1|.$$

Theorem [4.6]
(unitPhaseDistance_pos_of_mem_Ico_ne)
Let $\theta, \rho \in [0, 1)$ and $\theta \neq \rho$. Then,

$$d(\theta, \rho) > 0.$$

Theorem [4.7]
(unitPhaseDistance_le_half_of_mem_Ico)
Let $\theta, \rho \in [0, 1)$. Then,

$$d(\theta, \rho) \leq \frac{1}{2}.$$

Theorem [4.8]
(unitPhaseDistance_cases)
If $d(\theta, \rho) \leq \epsilon$, then at least one of the following statements is true:

1. $|\theta - \rho| \leq \epsilon$,
2. $|\theta - \rho - 1| \leq \epsilon$,
3. $|\theta - \rho + 1| \leq \epsilon$.

Definition [4.11]
(qpeCircularPhaseWindow)

$$d\left(\theta, \frac{y}{2^m}\right) \leq \frac{k}{2^m}.$$

Definition [4.12]
(qpeCircularPhaseWindowOutcomes)

$$O_{m,k,\theta} = \left\{ y \in \mathbb{Z}_{2^m} \middle| d\left(\theta, \frac{y}{2^m}\right) \leq \frac{k}{2^m} \right\}.$$

Definition [4.13]
(qpeCircularPhaseWindowProbability)
Let Y be the event of measuring y from $IQ(m)\,|p(m,\theta)\rangle$.

$$Pr\{y \in O_{m,k,\theta} | Y\} = \sum_{y \in O_{m,k,\theta}} ||a(m,\theta,y)||^2.$$

Definition [4.14]
(qpeCircularPhaseWindowFailureOutcomes)

$$F_{m,k,\theta} = O_{m,k,\theta}^{\complement}.$$

Definition [4.15]
(qpeCircularPhaseWindowFailureProbability)
Let Y be the event of measuring y from $IQ(m)\,|p(m,\theta)\rangle$.

$$Pr\{y \in F_{m,k,\theta} | Y\} = \sum_{y \in F_{m,k,\theta}} ||a(m,\theta,y)||^2.$$

Theorem [4.9]
(mem_qpeCircularPhaseWindowOutcomes_iff)
Let $m, k \in \mathbb{N}$, $\theta \in \mathbb{R}$, $y \in \mathbb{Z}_{2^m}$. Then, $y \in O_{m,k,\theta}$ if and only if $d\left(\theta, \frac{y}{2^m}\right) \leq \frac{k}{2^m}$.

Theorem [4.10]
(mem_qpeCircularPhaseWindowFailureOutcomes_iff)
Let $m, k \in \mathbb{N}$, $\theta \in \mathbb{R}$, $y \in \mathbb{Z}_{2^m}$. Then, $y \in F_{m,k,\theta}$ if and only if $\neg\left(d\left(\theta, \frac{y}{2^m}\right) \leq \frac{k}{2^m}\right)$.

Definition [4.16]
(floorGridIndexNat)

$$\lfloor 2^m \theta \rfloor.$$

Definition [4.17]
(qpeLowerAdjacentOutcome)

$$\lfloor 2^m \theta \rfloor \pmod{2^m}.$$

Definition [4.18]
(qpeUpperAdjacentOutcome)

$$\lfloor 2^m \theta \rfloor + 1 \pmod{2^m}.$$

Definition [4.19]
(qpeFractionalOffset)

$$2^m \theta - \lfloor 2^m \theta \rfloor.$$

Theorem [4.11]
(floorGridIndexNat_lt_M_of_mem_Ico)
Let $m \in \mathbb{N}$, $\theta \in [0, 1)$. Then,

$$\lfloor 2^m \theta \rfloor < 2^m.$$

Theorem [4.12]
(theta_eq_floorGrid_div_of_qpeFractionalOffset_eq_zero)
Let $m \in \mathbb{N}$, $\theta \geq 0$ and $2^m \theta - \lfloor 2^m \theta \rfloor = 0$. Then,

$$\theta = \frac{\lfloor 2^m \theta \rfloor}{2^m}.$$

Theorem [4.13]
(qpeLowerAdjacentOutcome_mem_circularWindow_one_of_mem_Ico)
Let $m \in \mathbb{N}$, $\theta \in [0, 1)$. Then,

$$\lfloor 2^m \theta \rfloor \in O_{m,1,\theta}.$$

Theorem [4.14]
(qpeUpperAdjacentOutcome_mem_circularWindow_one_of_mem_Ico)
Let $m \in \mathbb{N}$, $\theta \in [0,1)$. Then,

$$\lfloor 2^m\theta \rfloor + 1 \pmod{2^m} \in O_{m,1,\theta}.$$

Theorem [4.15]
(qpeAdjacentOutcomes_ne_of_pos_m)
Let $m \in \mathbb{N}$, $\theta \in [0,1)$, and $m > 0$. Then,

$$\lfloor 2^m\theta \rfloor \neq \lfloor 2^m\theta \rfloor + 1 \pmod{2^m}.$$

Theorem [4.16]
(qpeCircularFailure_unitPhaseDistance_gt)
Let $m, k \in \mathbb{N}$, $\theta \in \mathbb{R}$, and $y \in F_{m,k,\theta}$. Then,

$$d\left(\theta, \frac{y}{2^m}\right) > \frac{k}{2^m}.$$

Theorem [4.17]
(qpeCircularFailure_phase_error_representative)
Let $m \in \mathbb{N}$, $\theta \in [0,1]$, $k \in \mathbb{N}$, and $y \in F_{m,k,\theta}$. There exists δ such that at least one of the following statements is true:

1. $\delta = \theta - \frac{y}{2^m}$,
2. $\delta = \theta - \frac{y}{2^m} - 1$,
3. $\delta = \theta - \frac{y}{2^m} + 1$.

Moreover,

$$\frac{k}{2^m} < |\delta| \leq \frac{1}{2}.$$

Theorem [4.18]
(fin_eq_left_or_right_candidate_of_abs_sub_mem_Ico)
Let $N \in \mathbb{N}$ and $N > 0$. Let $x \in \mathbb{R}$, $i \in \mathbb{N}$, $y \in \mathbb{Z}_N$, and $z \in \mathbb{Z}$. Assume:

1. $y = z \pmod{N}$,
2. $i \le |x - z| < i + 1$.

Then, at least one of the following holds:

$$y = \lfloor x \rfloor - i \pmod{N},$$
$$y = \lceil x \rceil + i \pmod{N}.$$

Definition [4.20]
(qpeCircularDistanceBucket)

$$\left\lfloor 2^m d\left(\theta, \frac{y}{2^m}\right)\right\rfloor.$$

Theorem [4.19]
(qpeCircularDistanceBucket_mem_tail_of_failure)
Let $m, k \in \mathbb{N}$, and $\theta \in [0,1]$. Let $k > 0$ and $y \in F_{m,k,\theta}$. Then,

$$k - 1 < \left\lfloor 2^m d\left(\theta, \frac{y}{2^m}\right)\right\rfloor \le 2^m.$$

Theorem [4.20]
(qpeCircularFailure_phase_error_representative_with_unitDistance)
Let $m \in \mathbb{N}$, $\theta \in [0,1]$, $k \in \mathbb{N}$, and $y \in F_{m,k,\theta}$. There exists δ such that at least one of the following statements is true:

1. $\delta = \theta - \frac{y}{2^m}$,
2. $\delta = \theta - \frac{y}{2^m} - 1$,
3. $\delta = \theta - \frac{y}{2^m} + 1$.

Moreover,

$$\frac{k}{2^m} < |\delta| \le \frac{1}{2},$$

and

$$d\left(\theta, \frac{y}{2^m}\right) \le |\delta|.$$

Definition [4.21]
(bhmt11SuccessProbability)
If $k = 1$, then

$$\frac{8}{\pi^2}.$$

If $k \neq 1$, then,

$$1 - \frac{1}{2(k-1)}.$$

Definition [4.22]
(BHMT11CircularWindowBound)
Let $m, k \in \mathbb{N}$, $\theta \in [0,1]$. Assume also that $k > 0$.
If $k = 1$,

$$\frac{8}{\pi^2} \leq Pr\{y \in O_{m,k,\theta}|Y\}.$$

If $k > 1$,

$$1 - \frac{1}{2(k-1)} \leq Pr\{y \in O_{m,k,\theta}|Y\}.$$

Theorem [4.21]
(phaseState_add_one)
Let $m \in \mathbb{N}$ and $\theta \in \mathbb{R}$. Then,

$$|p(m, \theta + 1)\rangle = |p(m, \theta)\rangle .$$

Theorem [4.22]
(qpeApproxAmplitude_add_one)
Let $m \in \mathbb{N}$, $\theta \in \mathbb{R}$, and $y \in \mathbb{Z}_{2^m}$. Then,

$$a(m, \theta + 1, y) = a(m, \theta, y).$$

Theorem [4.23]
(qpeApproxOutcomeProbability_add_one)
Let $m \in \mathbb{N}$, $\theta \in \mathbb{R}$, and $y \in \mathbb{Z}_{2^m}$. Then,

$$||a(m, \theta + 1, y)||^2 = ||a(m, \theta, y)||^2.$$

Theorem [4.24]
(qpeApproxOutcomeProbability_one_sub)
Let $m \in \mathbb{N}$*,* $\theta \in \mathbb{R}$*, and* $y \in \mathbb{Z}_{2^m}$*. Then,*

$$||a(m, 1 - \theta, y)||^2 = ||a(m, -\theta, y)||^2.$$

Definition [4.23]
(qpeApproxGeometricProbability)
If $\delta = 0$*, then*

$$1.$$

If $\delta \neq 0$*, then*

$$\frac{\sin^2(N\pi\delta)}{N^2 \sin^2(\pi\delta)}.$$

Theorem [4.25]
(qpeCircularFailureProbability_le_of_bucket_tail)
Let $m, k, n \in \mathbb{N}$ *and* $k > 1$*. Let* $f : \mathbb{Z}_{2^m} \to \mathbb{N}$ *such that the following propositions hold:*

1. *For all* $y \in F_{m,k,\theta}$*,*

$$k - 1 < f(y) \leq n.$$

2. *For all* $y \in F_{m,k,\theta}$*,*

$$||a(m, \theta, y)||^2 \leq \frac{1}{4f(y)^2}.$$

3. *For all* $i \in (k - 1, n] \cap \mathbb{N}$*,*

$$|F_{m,k,\theta} \cap \{y \in \mathbb{Z}_{2^m} | f(y) = i\}| \leq 2.$$

Then,

$$Pr\{y \in F_{m,k,\theta} | Y\} \leq \frac{1}{2(k-1)}.$$

Definition [4.24]
(bhmtK1TwoNearestCore)

$$\frac{\sin^2(\pi x)}{x^2} + \frac{\sin^2(\pi x)}{(1-x)^2}.$$

Theorem [4.26]
(bhmtK1TwoNearestCore_symm)
bhmtK1TwoNearestCore *is symmetric over* $\frac{1}{2}$.

Theorem [4.27]
(bhmtK1TwoNearestCore_half)
bhmtK1TwoNearestCore *is* 8 *when* $x = \frac{1}{2}$.

Theorem [4.28]
(bhmtK1TwoNearestCore_ge_eight_of_antitone_left)
If bhmtK1TwoNearestCore *is non-increasing on* $\left(0, \frac{1}{2}\right)$ *and* $x \in (0,1)$, *then*

$$\frac{\sin^2(\pi x)}{x^2} + \frac{\sin^2(\pi x)}{(1-x)^2} \geq 8.$$

Theorem [4.29]
(deriv_bhmtK1TwoNearestCore)
Let $x \in \mathbb{R}$ *and* $x \neq 0, 1$. *Then,*

$$\frac{\mathrm{d}d}{\mathrm{d}dx}\left(\frac{\sin^2(\pi x)}{x^2} + \frac{\sin^2(\pi x)}{(1-x)^2}\right)(x) = \frac{\pi\sin(2\pi x)}{x^2} - \frac{2\sin^2(\pi x)}{x^3} + \frac{\pi\sin(2\pi x)}{(1-x)^2} + \frac{2\sin^2(\pi x)}{(1-x)^3}.$$

Theorem [4.30]
(bhmtK1TwoNearestCore_antitone_left_of_deriv_nonpos)
If the derivative of bhmtK1TwoNearestCore *is non-positive on* $\left(0, \frac{1}{2}\right)$, *then* bhmtK1TwoNearestCore *is non-increasing on* $\left(0, \frac{1}{2}\right)$.

Theorem [4.31]
(qpeApproxGeometricProbability_neg)
qpeApproxGeometricProbability *is symmetric on* δ *over* 0.

Theorem [4.32]
(qpeApproxGeometricProbability_abs)
qpeApproxGeometricProbability *is symmetric on* δ *over* 0.

Theorem [4.33] (qpeApproxGeometricProbability_add_one_of_sin_ne_zero)
Let $N \in \mathbb{N}$ and $\delta \in \mathbb{R}$. Assume that

$$\sin(\pi\delta) \neq 0.$$

Then,

$$\frac{\sin^2(N\pi(\delta+1))}{N^2\sin^2(\pi\delta)} = \frac{\sin^2(N\pi\delta)}{N^2\sin^2(\pi\delta)}.$$

Theorem [4.34] (qpeApproxGeometricProbability_sub_one_of_sin_ne_zero)
Let $N \in \mathbb{N}$ and $\delta \in \mathbb{R}$. Assume that

$$\sin(\pi\delta) \neq 0.$$

Then,

$$\frac{\sin^2(N\pi(\delta-1))}{N^2\sin^2(\pi\delta)} = \frac{\sin^2(N\pi\delta)}{N^2\sin^2(\pi\delta)}.$$

Theorem [4.35] (qpeApproxGeometricProbability_unitPhaseDistance_eq_of_sin_ne_zero)
Let $N \in \mathbb{N}$ and $\theta, \rho \in \mathbb{R}$ such that

$$\sin(\pi(\theta-\rho)) \neq 0.$$

Then,

$$\frac{\sin^2(N\pi d(\theta,\rho))}{N^2\sin^2(\pi d(\theta,\rho))} = \frac{\sin^2(N\pi(\theta-\rho))}{N^2\sin^2(\pi(\theta-\rho))}.$$

Theorem [4.36] (qpeApproxGeometricProbability_lower_bound_scaled)
Let $N \in \mathbb{N}$ and $x \in (0,1)$, $N > 0$, and $\delta = \frac{x}{N}$. Then,

$$\frac{\sin^2(\pi x)}{\pi^2 x^2} \leq \frac{\sin^2(N\pi\delta)}{N^2\sin^2(\pi\delta)}.$$

Theorem [4.37]
(qpeApproxGeometricProbability_two_nearest_lower_bound_of_core)
Let $N \in \mathbb{N}$ and $x \in (0,1)$, and $N > 0$. Then,

$$\frac{1}{\pi^2}\left(\frac{\sin^2(\pi x)}{x^2} + \frac{\sin^2(\pi x)}{(1-x)^2}\right) \leq \frac{\sin^2(\pi x)}{N^2 \sin^2\left(\pi \frac{x}{N}\right)} + \frac{\sin^2(\pi(x-1))}{N^2 \sin^2\left(\pi \frac{x-1}{N}\right)}.$$

Theorem [4.38]
(qpeCircularPhaseWindowProbability_lower_bound_two_outcomes)
Let $m, k \in \mathbb{N}$, $\theta, c \in \mathbb{R}$, $y_0, y_1 \in \mathbb{Z}_{2^m}$. Let also $y_0, y_1 \in O_{m,k,\theta}$ and $y_0 \neq y_1$. Assume the following.

$$c \leq ||a(m,\theta,y_0)||^2 + ||a(m,\theta,y_1)||^2.$$

Then,

$$c \leq Pr\{y \in O_{m,k,\theta} | Y\}.$$

Definition [4.25]
(controlledPowerMatrix)
Let U be a square matrix of dimension n, and $m \in \mathbb{N}$.

$$\Lambda(U,m) = \operatorname{diag}\left(I, U, \ldots, U^{2^m-1}\right).$$

Theorem [4.39]
(qftMatrix_mul_basisState)
Let $m \in \mathbb{N}$ and $y \in \mathbb{Z}_{2^m}$. Then,

$$Q(m)\,|y\rangle = |p\left(m, \frac{y}{2^m}\right)\rangle$$

Theorem [4.40]
(controlledPowerMatrix_mul_uniform_of_real_power_action)
Let $n, m \in \mathbb{N}$. Let U be a square matrix of dimension n, ψ a vector of dimension n, and $\theta \in \mathbb{R}$. Assume that, for all $k \in \mathbb{Z}_{2^m}$, the following holds.

$$U^k \psi = e^{2\pi i k\theta}\psi.$$

Then,

$$\Lambda(U, m)(|u(m)\rangle \otimes \psi) = |p(m, \theta)\rangle \otimes \psi.$$

Theorem [4.41]
(controlledPowerMatrix_mul_uniform_of_real_eigenphase)
Let $n, m \in \mathbb{N}$. Let U be a square matrix of dimension n, ψ a vector of dimension n, and $\theta \in \mathbb{R}$. Assume also that

$$U\psi = e^{2\pi i\theta}\psi.$$

Then,

$$\Lambda(U, m)(|u(m)\rangle \otimes \psi) = |p(m, \theta)\rangle \otimes \psi.$$

Theorem [4.42]
(phaseState_isNormalized)
Let $m \in \mathbb{N}$ and $\theta \in \mathbb{R}$. Then, $|p(m, \theta)\rangle$ is normalized.

Theorem [4.43]
(inverseQFTMatrix_eq_adjoint_qftMatrix)
Let $m \in \mathbb{N}$. Then,

$$IQ(m) = Q^{\dagger}(m)$$

Theorem [4.44]
(qft_phase_orthogonality)
Let $m \in \mathbb{N}$ and $r, c \in \mathbb{Z}_{2^m}$. Then,

$$\sum_{k=0}^{2^m-1} e^{2\pi ikr/2^m}{}^{\dagger} e^{2\pi ikc/2^m} = \begin{cases} 2^m & r = c \\ 0 & \textit{otherwise} \end{cases}.$$

Theorem [4.45]
(qft_normalization_factor)
Let $m \in \mathbb{N}$. Then,

$$\frac{1}{\sqrt{2^m}}\frac{1}{\sqrt{2^m}} = \frac{1}{2^m}.$$

Theorem [4.46]
(qpeApproxAmplitude_eq_normalized_phase_sum)
Let $m \in \mathbb{N}$, $\theta \in \mathbb{R}$, and $y \in \mathbb{Z}_{2^m}$. Then,

$$a(m, \theta, y) = \frac{1}{2^m} \sum_{k=0}^{2^m - 1} e^{2\pi i k\left(\theta - \frac{y}{2^m}\right)}.$$

Theorem [4.47]
(phase_fin_sum_normSq)
Let $N \in \mathbb{N}$, $\delta \in \mathbb{R}$. Let also $\sin(\pi\delta) \neq 0$. Then,

$$|| \sum_{k=0}^{N-1} e^{2\pi i k\delta} ||^2 = \frac{\sin^2(\pi N\delta)}{\sin^2(\pi\delta)}.$$

Theorem [4.48]
(qftMatrix_adjoint_mul_self)
Let $m \in \mathbb{N}$. Then,

$$Q^\dagger(m) Q(m) = I.$$

Theorem [4.49]
(qftMatrix_isUnitary)
Let $m \in \mathbb{N}$. $Q(m)$ is unitary.

Theorem [4.50]
(inverseQFTMatrix_isUnitary)
Let $m \in \mathbb{N}$. $IQ(m)$ is unitary.

Theorem [4.51]
(inverseQFTMatrix_mul_phaseState)
Let $m \in \mathbb{N}$ and $y \in \mathbb{Z}_{2^m}$. Then,

$$IQ(m) \left| p\left(m, \frac{y}{2^m}\right) \right\rangle = |y\rangle .$$

Theorem [4.52]
(qpeApproxOutcomeProbability_total)

Let $m \in \mathbb{N}$ and $\theta \in \mathbb{R}$. Then,

$$\sum_{y=0}^{2^m-1} ||a(m,\theta,y)||^2 = 1.$$

Theorem [4.53] (qpeApproxOutcomeProbability_eq_geometric_of_sin_ne_zero)
Let $m \in \mathbb{N}$, $\theta \in \mathbb{R}$ and $y \in \mathbb{Z}_{2^m}$. Assume that

$$\sin\left(\pi\left(\theta - \frac{y}{2^m}\right)\right) \neq 0.$$

Then,

$$||a(m,\theta,y)||^2 = \frac{\sin^2(2^m\pi\left(\theta - \frac{y}{2^m}\right))}{2^{2m}\sin^2(\pi\left(\theta - \frac{y}{2^m}\right))}.$$

Theorem [4.54] (qpeApproxOutcomeProbability_eq_geometric_of_nearest)
Let $m \in \mathbb{N}$, $\theta \in \mathbb{R}$ and $y \in \mathbb{Z}_{2^m}$. Assume the following.

$$\left|\theta - \frac{y}{2^m}\right| \leq \frac{1}{2^{m+1}}.$$

If $\theta - \frac{y}{2^m} \neq 0$,

$$||a(m,\theta,y)||^2 = \frac{\sin^2(2^m\pi\left(\theta - \frac{y}{2^m}\right))}{2^{2m}\sin^2(\pi\left(\theta - \frac{y}{2^m}\right))}.$$

And otherwise,

$$||a(m,\theta,y)||^2 = 1.$$

Theorem [4.55] (qpeApproxGeometricProbability_upper_bound_sin_den)
Let $N \in \mathbb{N}$ and $\delta \in \mathbb{R}$. Assume that $N > 0$ and that the following holds.

$$\sin(\pi\delta) \neq 0.$$

Then,

$$\frac{\sin^2(N\pi\delta)}{N^2\sin^2(\pi\delta)} \leq \frac{1}{N^2\sin^2(\pi\delta)}.$$

Theorem [4.56] (qpeApproxGeometricProbability_upper_bound_unitDistance)
Let $N \in \mathbb{N}$ and $\delta \in (0, \frac{1}{2}]$. Assume that $N > 0$. Then,

$$\frac{\sin^2(N\pi\delta)}{N^2\sin^2(\pi\delta)} \leq \frac{1}{4N^2\delta^2}.$$

Theorem [4.57] (qpeApproxGeometricProbability_upper_bound_of_sin_lower)
Let $N, j \in \mathbb{N}$ and $\delta \in \mathbb{R}$. Let also $N, j > 0$, $\sin(\pi\delta) \neq 0$ and

$$\frac{2j}{N} \leq |\sin(\pi\delta)|.$$

Then,

$$\frac{\sin^2(N\pi\delta)}{N^2\sin^2(\pi\delta)} \leq \frac{1}{4j^2}.$$

Corollary (qpeApproxOutcomeProbability_upper_bound_of_sin_lower)
The same theorem holds for $\delta = \theta - \frac{y}{2^m}$, for all $\theta \in \mathbb{R}$ and $y \in \mathbb{Z}_{2^m}$.

Theorem [4.58] (qpeApproxOutcomeProbability_failure_upper_bound_distanceBucket)
Let $m, k \in \mathbb{N}$, $\theta \in [0, 1]$, and $y \in \mathbb{Z}_{2^m}$. Let $k > 1$ and $y \in F_{m,k,\theta}$. Then,

$$||a(m, \theta, y)||^2 \leq \frac{1}{4\lfloor 2^m d\left(\theta, \frac{y}{2^m}\right)\rfloor^2}.$$

Theorem [4.59] (qpeCircularDistanceBucket_eq_imp_left_or_right_candidate)
Let $m, i \in \mathbb{N}$, $\theta \in \mathbb{R}$, and $y \in \mathbb{Z}_{2^m}$. Assume that

$$\lfloor 2^m d\left(\theta, \frac{y}{2^m}\right)\rfloor = i.$$

Then, at least one of the following two statements holds.

1. $y = \lfloor 2^m\theta\rfloor - i \pmod{2^m}$,
2. $y = \lceil 2^m\theta\rceil + i \pmod{2^m}$.

Theorem [4.60] (qpeCircularDistanceBucket_filter_card_le_two)
Let $m, k, i \in \mathbb{N}$ and $\theta \in \mathbb{R}$. Then,

$$|F_{m,k,\theta} \cap \{y | \lfloor 2^m d\left(\theta, \frac{y}{2^m}\right)\rfloor = i\}| \leq 2.$$

Theorem [4.61] (qpeCircularFailureProbability_le_k_gt_one)
Let $m, k \in \mathbb{N}$ and $\theta \in [0, 1]$. Let $k > 1$. Then,

$$Pr\{y \in F_{m,k,\theta} | Y\} \leq \frac{1}{2(k-1)}.$$

Theorem [4.62] (qpeCircularPhaseWindowProbability_lower_bound_k_gt_one)
Let $m, k \in \mathbb{N}$ and $\theta \in [0, 1]$. Let $k > 1$. Then,

$$1 - \frac{1}{2(k-1)} \leq Pr\{y \in O_{m,k,\theta} | Y\}.$$

Theorem [4.63] (qpeApproxOutcomeProbability_lowerAdjacent_eq_one_of_fractionalOffset_eq_zero)
Let $m \in \mathbb{N}$ and $\theta \in [0, 1)$. Let also

$$2^m\theta - \lfloor 2^m\theta\rfloor = 0.$$

Then,

$$||a(m, \theta, 2^m\theta)||^2 = 1$$

Theorem [4.64]
(qpeCircularPhaseWindowProbability_lower_bound_k1_exactGrid_of_mem_Ico)
Let $m \in \mathbb{N}$ and $\theta \in [0, 1)$. Let also

$$2^m\theta - \lfloor 2^m\theta \rfloor = 0.$$

Then,

$$\frac{8}{\pi^2} \leq Pr\{y \in O_{m,1,\theta} | Y\}.$$

Theorem [4.65]
(qpeApproxOutcomeProbability_lowerAdjacent_eq_geometric_fractionalOffset)
Let $m \in \mathbb{N}$ and $\theta \in [0, 1)$, and $\delta = \frac{2^m\theta - \lfloor 2^m\theta \rfloor}{2^m}$. Assume that

$$2^m\theta - \lfloor 2^m\theta \rfloor > 0.$$

Then,

$$||a(m, \theta, \lfloor 2^m\theta \rfloor)||^2 = \frac{\sin^2(2^m\pi\delta)}{2^{2m}\sin^2(\pi\delta)}.$$

Theorem [4.66]
(qpeApproxOutcomeProbability_upperAdjacent_eq_geometric_fractionalOffset)
Let $m \in \mathbb{N}$ and $\theta \in [0, 1)$, and $\delta = \frac{2^m\theta - \lfloor 2^m\theta \rfloor}{2^m} - \frac{1}{2^m}$. Assume that

$$2^m\theta - \lfloor 2^m\theta \rfloor > 0.$$

If $\delta \neq 0$, then

$$||a(m, \theta, \lfloor 2^m\theta \rfloor + 1 \pmod{2^m})||^2 = \frac{\sin^2(2^m\pi\delta)}{2^{2m}\sin^2(\pi\delta)}.$$

And if $\delta = 0$, then

$$||a(m, \theta, \lfloor 2^m\theta \rfloor + 1 \pmod{2^m})||^2 = 1.$$

5 QuantumLibraryBridge

Definition [5.1]
(qaePlaneVector)

$$\cos(\theta) |0\rangle + \sin(\theta) |1\rangle .$$

6 GroverQPEBridge

Definition [6.1]
(ketMinusI)

$$|{-i}\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i |1\rangle).$$

Definition [6.2]
(ketPlusI)

$$|{+i}\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i |1\rangle).$$

Theorem [6.1]
(Ry_ketMinusI_eigen)
Let $\theta \in \mathbb{R}$. Then,

$$R_y(4\theta) |{-i}\rangle = e^{2i\theta} |{-i}\rangle .$$

Theorem [6.2]
(Ry_ketPlusI_eigen)
Let $\theta \in \mathbb{R}$. Then,

$$R_y(4\theta) |{+i}\rangle = e^{-2i\theta} |{+i}\rangle .$$

7 GroverQAESuperposition

Definition [7.1]
(qaeCoeffPlus)

$$\frac{1}{\sqrt{2}}(\cos(\theta) + i\sin(\theta)).$$

Definition [7.2]
(qaeCoeffMinus)

$$\frac{1}{\sqrt{2}}(\cos(\theta) - i\sin(\theta)).$$

Theorem [7.1]
(qaeCoeffPlus_normSq)

$$||\frac{1}{\sqrt{2}}(\cos(\theta) + i\sin(\theta))||^2 = \frac{1}{2}.$$

Theorem [7.2]
(qaeCoeffMinus_normSq)

$$||\frac{1}{\sqrt{2}}(\cos(\theta) - i\sin(\theta))||^2 = \frac{1}{2}.$$

Theorem [7.3]
(qaePlaneVector_eq_eigen_superposition)

$$\cos(\theta)\left|0\right\rangle + \sin(\theta)\left|1\right\rangle = \frac{1}{\sqrt{2}}(\cos(\theta) + i\sin(\theta))\left|-i\right\rangle + \frac{1}{\sqrt{2}}(\cos(\theta) - i\sin(\theta))\left|+i\right\rangle.$$

Theorem [7.4]
(qpeOutputStateConcrete_linear_combination)
Let $n, m \in \mathbb{N}$, U a square matrix of dimension n, ψ, ϕ vectors of dimension n, and $a, b \in \mathbb{C}$. Then,

$$
\begin{aligned}
&(IQ(m) \otimes I_n) \times \Lambda(U, m) \times (|u(m)\rangle \otimes (a\psi + b\phi)) \\
&= a(IQ(m) \otimes I_n) \times \Lambda(U, m) \times (|u(m)\rangle \otimes \psi) \\
&+ b(IQ(m) \otimes I_n) \times \Lambda(U, m) \times (|u(m)\rangle \otimes \phi).
\end{aligned}
$$

Theorem [7.5]
(qpeOutput_groverPlane_initial_eq_eigenphase_superposition)
Let $m \in \mathbb{N}$ *and* $\theta \in \mathbb{R}$. *Then,*

$$
\begin{aligned}
&(IQ(m) \otimes I_2) \times \Lambda(R_y(4\theta), m) \times (|u(m)\rangle \otimes (\cos(\theta)\,|0\rangle + \sin(\theta)\,|1\rangle)) \\
&= \frac{1}{\sqrt{2}}(\cos(\theta) + i\sin(\theta)) IQ(m) \times (|p(m, \frac{\theta}{\pi})\rangle \otimes |{-i}\rangle) \\
&+ \frac{1}{\sqrt{2}}(\cos(\theta) - i\sin(\theta)) IQ(m) \times (|p(m, -\frac{\theta}{\pi})\rangle \otimes |{+i}\rangle).
\end{aligned}
$$

Definition [7.3]
(qpeCountingMarginal)
Let $n, m \in \mathbb{N}$, ψ *a vector of dimension* $2^m \times n$, *and* $y \in \mathbb{Z}_{2^m}$.

$$
\sum_{j=0}^{n-1} ||\psi_{j+ny}||^2.
$$

Theorem [7.6]
(qpeCountingMarginal_groverPlane_initial)
Let $m \in \mathbb{N}$, $\theta \in \mathbb{R}$, *and* $y \in \mathbb{Z}_{2^m}$. *Let also*

$$
v = (IQ(m) \otimes I_2) \times \Lambda(R_y(4\theta), m) \times (|u(m)\rangle \otimes (\cos(\theta)\,|0\rangle + \sin(\theta)\,|1\rangle)).
$$

Then,

$$
\sum_{j=0}^{1} ||v_{j+2y}||^2 = \frac{1}{2}||a(m, \frac{\theta}{\pi}, y)||^2 + \frac{1}{2}||a(m, -\frac{\theta}{\pi}, y)||^2.
$$

8 QuantumAmplitudeEstimation

Definition [8.1]
(amplitudeFromAngle)
Let $\theta \in \mathbb{R}$.

$$\sin^2(\theta).$$

Definition [8.2]
(estAmpEstimate)
Let $M, y \in \mathbb{N}$.

$$\sin^2\left(\pi \frac{y}{M}\right).$$

Definition [8.3]
(phaseErrorRadius)
Let $M, k \in \mathbb{N}$.

$$\pi \frac{k}{M}.$$

Definition [8.4]
(theorem12ErrorBound)
Let $a \in \mathbb{R}$ and $M, k \in \mathbb{N}$.

$$2\pi \frac{k}{M}\sqrt{a(1-a)} + \pi^2 \frac{k^2}{M^2}.$$

Definition [8.5]
(theorem12SuccessProbability)
Let $k \in \mathbb{N}$. If $k = 1$, then

$$\frac{8}{\pi^2}.$$

If $k \neq 1$, then

$$1 - \frac{1}{2(k-1)}.$$

Theorem [8.1]
(paperLemma7)
Let $a, \theta, \hat{\theta}, \epsilon$. *Let also* $\epsilon \geq 0$, $a = \sin^2(\theta)$. *Assume that*

$$|\theta - \hat{\theta}| \leq \epsilon.$$

Then,

$$|\sin^2(\hat{\theta}) - a| \leq 2\epsilon\sqrt{a(1-a)} + \epsilon^2.$$

Theorem [8.2]
(PaperTheorem11)
Let $m, k \in \mathbb{N}$ *and* $\theta \in [0, 1]$. *The following four statements hold:*

1. *If* $2^m\theta - \lfloor 2^m\theta \rfloor = 0$ *and* $\theta \neq 1$, *then*

$$||a(m, \theta, 2^m\theta)||^2 = 1.$$

2. *If* $2^m\theta - \lfloor 2^m\theta \rfloor \neq 0$, *let* $x \in \mathbb{Z}_{2^m}$, $\theta \neq 1$, *and* $\delta = d(\theta, \frac{x}{2^m})$. *Then,*

$$||a(m, \theta, x)||^2 = \frac{\sin^2(2^m\pi\delta)}{2^{2m}\sin^2(\pi\delta)} \leq \frac{1}{2^{2m+2}\delta^2}.$$

3. *If* $k > 1$, *then*

$$Pr\{y \in O_{m,k,\theta}|Y\} \geq 1 - \frac{1}{2(k-1)}.$$

4. *If* $k = 1$ *and* $2^m > 2$, *then*

$$Pr\{y \in O_{m,k,\theta}|Y\} \geq \frac{8}{\pi^2}.$$

Definition [8.6]
(phaseOracle)
Let $n \in \mathbb{N}$ *and* $f : \mathbb{Z}_{2^n} \to \{0, 1\}$.

$$S_f^n |k\rangle = \begin{cases} -|k\rangle, & f(k) = 1 \\ |k\rangle, & f(k) = 0 \end{cases}.$$

Definition [8.7]
(zeroReflection)
Let $n \in \mathbb{N}$.

$$S_0^n |k\rangle = \begin{cases} -|k\rangle, & k = 0 \\ |k\rangle, & \textit{otherwise} \end{cases}.$$

Theorem [8.3]
(zeroReflection_mul_zero_basis)
Let $n \in \mathbb{N}$. *Then,*

$$S_0^n |0\rangle = -|0\rangle. \tag{1}$$

Definition [8.8]
(paperGroverOperator)
Let $n \in \mathbb{N}$, A *a square matrix of dimension* 2^n, *and* $f : \mathbb{Z}_{2^n} \to \{0, 1\}$.

$$G_f(A) = -AS_0^n A^\dagger S_f^n.$$

Theorem [8.4]
(zeroReflection_mul_vector)
Let $n \in \mathbb{N}$ *and* v *be a vector of dimension* 2^n. *Then,*

$$S_0^n v = v - 2v_0 |0\rangle.$$

Theorem [8.5]
(conjugatedZeroReflection_mul_vector)

Let $n \in \mathbb{N}$, A a unitary matrix of dimension 2^n, and v a vector of dimension 2^n. Then,

$$AS_0^n A^\dagger v = v - 2(A^\dagger v)_0 A |0\rangle .$$

Theorem [8.6]
(phaseOracle_mul_of_bad_support)
Let $n \in \mathbb{N}$, $f : \mathbb{Z}_{2^n} \to \{0,1\}$, and v a vector of dimension 2^n. Assume that $v_k = 0$ holds for all k such that $f(k) = 1$. Then,

$$S_f^n v = v.$$

Theorem [8.7]
(phaseOracle_mul_of_good_support)
Let $n \in \mathbb{N}$, $f : \mathbb{Z}_{2^n} \to \{0,1\}$, and v a vector of dimension 2^n. Assume that $v_k = 0$ holds for all k such that $f(k) = 0$. Then,

$$S_f^n v = -v.$$

Theorem [8.8]
(bad_good_support_orthogonal)
Let $n \in \mathbb{N}$, $f : \mathbb{Z}_{2^n} \to \{0,1\}$, and ψ_0, ψ_1 be vector of dimension 2^n. Assume that $\psi_{0k} = 0$ for all k such that $f(k) = 1$, while $\psi_{1k} = 0$ for all k such that $f(k) = 0$. Then,

$$\langle \psi_0 | \psi_1 \rangle = 0.$$

Corollary
(good_bad_support_orthogonal)
Also,

$$\langle \psi_1 | \psi_0 \rangle = 0.$$

Theorem [8.9]
(badComponent_overlap_of_decomposition)

Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n, ψ_0, ψ_1 vectors of dimension 2^n, and $\theta \in \mathbb{R}$. Assume that

$$A|0\rangle = \cos(\theta)\psi_0 + \sin(\theta)\psi_1.$$

Assume also that ψ_0 is normalized and that $\langle\psi_1|\psi_0\rangle = 0$. Then,

$$(A^\dagger\psi_0)_0 = \cos\theta.$$

Theorem [8.10] (goodComponent_overlap_of_decomposition)
Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n, ψ_0, ψ_1 vectors of dimension 2^n, and $\theta \in \mathbb{R}$. Assume that

$$A|0\rangle = \cos(\theta)\psi_0 + \sin(\theta)\psi_1.$$

Assume also that ψ_1 is normalized and that $\langle\psi_0|\psi_1\rangle = 0$. Then,

$$(A^\dagger\psi_1)_0 = \sin\theta.$$

Theorem [8.11] (paperGroverOperator_mul_badComponent)
Let $n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n, ψ_0, ψ_1 vectors of dimension 2^n, $f : \mathbb{Z}_{2^n} \to \{0, 1\}$, and $\theta \in \mathbb{R}$. Assume that $\psi_{0k} = 0$ for all k such that $f(k) = 1$, and that

$$A|0\rangle = \cos(\theta)\psi_0 + \sin(\theta)\psi_1.$$

Assume also that

$$(A^\dagger\psi_0)_0 = \cos(\theta).$$

Then,

$$G_f(A)\psi_0 = \cos(2\theta)\psi_0 + \sin(2\theta)\psi_1.$$

Theorem [8.12]
(paperGroverOperator_mul_goodComponent)
Let $n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n, ψ_0, ψ_1 vectors of dimension 2^n, $f : \mathbb{Z}_{2^n} \to \{0, 1\}$, and $\theta \in \mathbb{R}$. Assume that $\psi_{1_k} = 0$ for all k such that $f(k) = 0$, and that

$$A\,|0\rangle = \cos(\theta)\psi_0 + \sin(\theta)\psi_1.$$

Assume also that

$$(A^\dagger \psi_1)_0 = \sin(\theta).$$

Then,

$$G_f(A)\psi_1 = -\sin(2\theta)\psi_0 + \cos(2\theta)\psi_1.$$

Theorem [8.13]
(groverRotation_eigenMinus)
Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n, ψ_0, ψ_1 be vectors of dimension 2^n, and $\theta \in \mathbb{R}$. Assume that the following two propositions hold.

$$\begin{aligned} A\psi_0 &= \cos(2\theta)\psi_0 + \sin(2\theta)\psi_1, \\ A\psi_1 &= -\sin(2\theta)\psi_0 + \cos(2\theta)\psi_1. \end{aligned}$$

Then,

$$\frac{1}{\sqrt{2}}A(\psi_0 - i\psi_1) = \frac{1}{\sqrt{2}}e^{2\theta i}(\psi_0 - i\psi_1).$$

Theorem [8.14]
(groverRotation_eigenPlus)
Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n, ψ_0, ψ_1 be vectors of dimension 2^n, and $\theta \in \mathbb{R}$. Assume that the following two propositions hold.

$$\begin{aligned} A\psi_0 &= \cos(2\theta)\psi_0 + \sin(2\theta)\psi_1, \\ A\psi_1 &= -\sin(2\theta)\psi_0 + \cos(2\theta)\psi_1. \end{aligned}$$

Then,

$$\frac{1}{\sqrt{2}}A(\psi_0 + i\psi_1) = \frac{1}{\sqrt{2}}e^{-2\theta i}(\psi_0 + i\psi_1).$$

Theorem [8.15]
(paperGroverOperator_eigenvectors_of_good_bad_decomposition)
Let $n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n, ψ_0, ψ_1 vectors of dimension 2^n, $f : \mathbb{Z}_{2^n} \to \{0,1\}$, and $\theta \in \mathbb{R}$. Assume that $\psi_{0k} = 0$ for all k such that $f(k) = 1$, while $\psi_{1k} = 0$ for all k such that $f(k) = 0$. Assume that ψ_0 and ψ_1 are normalized. Assume also that

$$A\,|0\rangle = \cos(\theta)\psi_0 + \sin(\theta)\psi_1.$$

Then, the following two statements hold.

$$\frac{1}{\sqrt{2}}A(\psi_0 - i\psi_1) = \frac{1}{\sqrt{2}}e^{2\theta i}(\psi_0 - i\psi_1),$$
$$\frac{1}{\sqrt{2}}A(\psi_0 + i\psi_1) = \frac{1}{\sqrt{2}}e^{-2\theta i}(\psi_0 + i\psi_1).$$

Definition [8.9]
(paperQAEInitialState)
Let $m, n \in \mathbb{N}$ and A be a square matrix of dimension 2^n.

$$E_1(m,n,A) = \underbrace{|0\dots0\rangle}_{m} \otimes (A\,\underbrace{|0\dots0\rangle}_{n}).$$

Definition [8.10]
(paperQAEAfterQFT)
Let $m, n \in \mathbb{N}$ and A be a square matrix of dimension 2^n.

$$E_2(m,n,A) = (Q(m) \otimes I_n) \times E_1.$$

Definition [8.11]
(paperQAEAfterControlledPowers)
Let $m, n \in \mathbb{N}$, A be a square matrix of dimension 2^n, and $f : \mathbb{Z}_{2^n} \to \{0,1\}$.

$$E_3(m, n, A, f) = \Lambda(G_f(A), m) \times E_2.$$

Definition [8.12]
(paperQAEFinalState)
Let $m, n \in \mathbb{N}$, A *be a square matrix of dimension* 2^n, *and* $f : \mathbb{Z}_{2^n} \to \{0, 1\}$.

$$E_4(m, n, A, f) = (IQ(m) \otimes I_n) \times E_3.$$

Definition [8.13]
(paperQAEOutputProbability)
Let $m, n \in \mathbb{N}$, A *be a square matrix of dimension* 2^n, $f : \mathbb{Z}_{2^n} \to \{0, 1\}$, *and* $y \in \mathbb{Z}_{2^m}$.

$$\langle y| E_4(m, n, A, f) |y\rangle = \sum_{j=0}^{2^n-1} \langle y, j| E_4(m, n, A, f) |y, j\rangle .$$

Definition [8.14]
(paperQAEEstimate)

$$\sin^2 \left(\pi \frac{y}{2^m} \right) .$$

Definition [8.15]
(paperPreparedState)

$$A |0\rangle .$$

Definition [8.16]
(paperBadProjection)

$$b(n, A, f) = \sum_{\substack{k=0 \\ f(k)=0}}^{2^n-1} \langle k| A |0\rangle |k\rangle .$$

Definition [8.17]
(paperGoodProjection)

$$g(n, A, f) = \sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \langle k| A |0\rangle |k\rangle .$$

Definition [8.18]
(paperBadProbability)

$$\sum_{\substack{k=0 \\ f(k)=0}}^{2^n-1} \langle k| A |0\rangle \langle 0| A^\dagger |k\rangle .$$

Definition [8.19]
(paperGoodProbability)

$$\sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \langle k| A |0\rangle \langle 0| A^\dagger |k\rangle .$$

Definition [8.20]
(paperBadState)

$$\frac{1}{\cos(\theta)} \sum_{\substack{k=0 \\ f(k)=0}}^{2^n-1} \langle k| A |0\rangle |k\rangle .$$

Definition [8.21]
(paperGoodState)

$$\frac{1}{\sin(\theta)} \sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \langle k| A |0\rangle |k\rangle .$$

Theorem [8.16]
(paperPreparedState_decompose_bad_good)
Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n, and $f : \mathbb{Z}_{2^n} \to \{0,1\}$. Then,

$$A|0\rangle = \sum_{\substack{k=0 \\ f(k)=0}}^{2^n-1} \langle k| A |0\rangle |k\rangle + \sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \langle k| A |0\rangle |k\rangle .$$

Corollary
(paperPreparedState_decompose_bad_good_states)

$$A|0\rangle = \cos(\theta) \sum_{\substack{k=0 \\ f(k)=0}}^{2^n-1} \frac{1}{\cos(\theta)} \langle k| A |0\rangle |k\rangle + \sin(\theta) \sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \frac{1}{\sin(\theta)} \langle k| A |0\rangle |k\rangle .$$

Theorem [8.17]
(paperBadState_support)
Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n, $f : \mathbb{Z}_{2^n} \to \{0,1\}$, and $x \in \mathbb{Z}_{2^n}$. Assume that $f(x) = 1$. Then,

$$\left(\frac{1}{\cos(\theta)} \sum_{\substack{k=0 \\ f(k)=0}}^{2^n-1} \langle k| A |0\rangle |k\rangle \right)_x = 0.$$

Theorem [8.18]
(paperGoodState_support)
Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n, $f : \mathbb{Z}_{2^n} \to \{0,1\}$, and $x \in \mathbb{Z}_{2^n}$. Assume that $f(x) = 0$. Then,

$$\left(\frac{1}{\sin(\theta)} \sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \langle k| A |0\rangle |k\rangle \right)_x = 0.$$

Theorem [8.19]
(paperBadProjection_norm)

Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n, and $f : \mathbb{Z}_{2^n} \to \{0,1\}$. Then,

$$b(n,A,f)^\dagger b(n,A,f) = \sum_{\substack{k=0 \\ f(k)=0}}^{2^n-1} \langle k| A |0\rangle \langle 0| A^\dagger |k\rangle .$$

Theorem [8.20]
(paperGoodProjection_norm)
Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n, and $f : \mathbb{Z}_{2^n} \to \{0,1\}$. Then,

$$g(n,A,f)^\dagger g(n,A,f) = \sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \langle k| A |0\rangle \langle 0| A^\dagger |k\rangle .$$

Theorem [8.21]
(paperBadState_isNormalized)
Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n, $f : \mathbb{Z}_{2^n} \to \{0,1\}$, and $\theta \in \mathbb{R}$. Assume that

$$\sum_{\substack{k=0 \\ f(k)=0}}^{2^n-1} \langle k| A |0\rangle \langle 0| A^\dagger |k\rangle = 1 - \sin^2(\theta).$$

Also, assume that $\cos(\theta) \neq 0$. *Then,* $\frac{1}{\cos(\theta)} b(n,A,f)$ *is normalized.*

Theorem [8.22]
(paperGoodState_isNormalized)
Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n, $f : \mathbb{Z}_{2^n} \to \{0,1\}$, and $\theta \in \mathbb{R}$. Assume that

$$\sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \langle k| A |0\rangle \langle 0| A^\dagger |k\rangle = \sin^2(\theta).$$

Also, assume that $\sin(\theta) \neq 0$. *Then,* $\frac{1}{\sin(\theta)} g(n,A,f)$ *is normalized.*

Theorem [8.23]
(paperBadProbability_add_goodProbability)
Let $n \in \mathbb{N}$, A be a square matrix of dimension 2^n, and $f : \mathbb{Z}_{2^n} \to \{0,1\}$. Then,

$$\sum_{k=0}^{2^n-1} \left(\langle k| \, b(n, A, f) \, |k\rangle + \langle k| \, g(n, A, f) \, |k\rangle \right) = \sum_{k=0}^{2^n-1} \langle k| \, A \, |0\rangle \, \langle 0| \, A^\dagger \, |k\rangle \, .$$

Theorem [8.24]
(paperBadProbability_add_goodProbability_eq_one)
Let $n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n, and $f : \mathbb{Z}_{2^n} \to \{0,1\}$. Then,

$$\sum_{k=0}^{2^n-1} \left(\langle k| \, b(n, A, f) \, |k\rangle + \langle k| \, g(n, A, f) \, |k\rangle \right) = 1.$$

Corollary
(paperBadProbability_eq_one_sub_goodProbability)

$$\sum_{k=0}^{2^n-1} \langle k| \, b(n, A, f) \, |k\rangle = 1 - \sum_{k=0}^{2^n-1} \langle k| \, g(n, A, f) \, |k\rangle \, .$$

Theorem [8.25]
(paperBadProbability_zero_support)
Let $n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n, $f : \mathbb{Z}_{2^n} \to \{0,1\}$, and $x \in \mathbb{Z}_{2^n}$. Assume that $f(x) = 0$ and that

$$\sum_{\substack{k=0 \\ f(k)=0}}^{2^n-1} \langle k| \, A \, |0\rangle \, \langle 0| \, A^\dagger \, |k\rangle = 0.$$

Then,

$$(A \, |0\rangle)_x = 0.$$

Theorem [8.26]
(paperGoodProbability_zero_support)
Let $n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n, $f : \mathbb{Z}_{2^n} \to \{0, 1\}$, and $x \in \mathbb{Z}_{2^n}$. Assume that $f(x) = 1$ and that

$$\sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \langle k| A |0\rangle \langle 0| A^\dagger |k\rangle = 0.$$

Then,

$$(A |0\rangle)_x = 0.$$

Theorem [8.27]
(phaseState_zero_eq_uniformState)
Let $m \in \mathbb{N}$. Then,

$$Q(m) |p(m, 0)\rangle = |u(m)\rangle .$$

Theorem [8.28]
(paperQAEAfterQFT_eq_uniformState_tensor)
Let $m, n \in \mathbb{N}$ and A be a square matrix of dimension 2^n. Then,

$$E_2(m, n, A) = |u(m)\rangle \otimes (A \underbrace{|0 \ldots 0\rangle}_{n}).$$

Theorem [8.29]
(paperQAEFinalState_eq_eigenphase_superposition)
Let $m, n \in \mathbb{N}$, A be a square matrix of dimension 2^n, $f : \mathbb{Z}_{2^n} \to \{0, 1\}$, ψ, ϕ vectors of dimension 2^n, $\theta_\psi, \theta_\phi \in \mathbb{R}$, and $a, b \in \mathbb{C}$. Assume that

$$A |0\rangle = a\psi + b\phi.$$

Assume also that the following two propositions hold.

$$G_f(A)\psi = e^{2\pi i\theta_\psi}\psi,$$
$$G_f(A)\phi = e^{2\pi i\theta_\phi}\phi.$$

Then,

$$E_4(m,n,A,f) = aIQ(m)(|p(m,\theta_\psi)\rangle \otimes \psi) + bIQ(m)(|p(m,\theta_\phi)\rangle \otimes \phi).$$

Theorem [8.30]
(paperQAEOutputProbability_eq_partialProb)
Let $n \in \mathbb{N}$*,* A *be a square matrix of dimension* 2^n*,* $f : \mathbb{Z}_{2^n} \to \{0,1\}$*, and* $y \in \mathbb{Z}_{2^m}$*. Then,*

$$\langle y|\, E_4(m,n,A,f)\, |y\rangle = \sum_{j=0}^{2^n-1} \langle y,j|\, E_4(m,n,A,f)\, |y,j\rangle\,.$$

Theorem [8.31]
(paperQAEOutputProbability_eq_single_eigenphase)
Let $m,n \in \mathbb{N}$*,* A *be a unitary matrix of dimension* 2^n*,* $f : \mathbb{Z}_{2^n} \to \{0,1\}$*,* $y \in \mathbb{Z}_{2^m}$*, and* $\theta \in \mathbb{R}$*. Assume that*

$$G_f(A)A\,|0\rangle = e^{2\pi i\theta}A\,|0\rangle\,.$$

Then,

$$\langle y|\, E_4(m,n,A,f)\, |y\rangle = ||a(m,\theta,y)||^2.$$

Theorem [8.32]
(paperQAEOutputProbability_eq_eigenphase_mixture)
Let $m,n \in \mathbb{N}$*,* A *be a square matrix of dimension* 2^n*,* $f : \mathbb{Z}_{2^n} \to \{0,1\}$*,* ψ,ϕ *vectors of dimension* 2^n*,* $\theta_\psi,\theta_\phi \in \mathbb{R}$*, and* $a,b \in \mathbb{C}$*. Assume that*

$$A\,|0\rangle = a\psi + b\phi.$$

Assume that the following two propositions hold.

$$G_f(A)\psi = e^{2\pi i\theta_\psi}\psi,$$
$$G_f(A)\phi = e^{2\pi i\theta_\phi}\phi.$$

Also, assume that ψ *and* ϕ *are orthonormal. Let* $y \in \mathbb{Z}_{2^m}$*. Then,*

$$\langle y| E_4(m,n,A,f)|y\rangle = ||a||^2||a(m,\theta_\psi,y)||^2 + ||b||^2||a(m,\theta_\phi,y)||^2.$$

Theorem [8.33]
(goodBad_decomposition_eq_eigen_superposition)
Let $n \in \mathbb{N}$*,* ψ_0, ψ_1 *be vectors of dimension* 2^n*, and* $\theta \in \mathbb{R}$*. Then,*

$$\cos(\theta)\psi_0 + \sin(\theta)\psi_1 = \frac{1}{2}(cos(\theta) + i\sin(\theta))(\psi_0 - i\psi_1) + \frac{1}{2}(cos(\theta) - i\sin(\theta))(\psi_0 + i\psi_1).$$

Theorem [8.34]
(orthonormal_linear_combination_isNormalized)
Let $n \in \mathbb{N}$*,* ψ_0, ψ_1 *be vectors of dimension* 2^n*, and* $a, b \in \mathbb{C}$*. Assume that* ψ_0 *and* ψ_1 *are orthonormal. And that* $a^\dagger a + b^\dagger b = 1$*. Then,* $a\psi_0 + b\psi_1$ *is normalized.*

Theorem [8.35]
(orthonormal_linear_combination_inner)
Let $n \in \mathbb{N}$*,* ψ_0, ψ_1 *be vectors of dimension* 2^n*, and* $a, b, c, d \in \mathbb{C}$*. Assume that* ψ_0 *and* ψ_1 *are orthonormal. Then,*

$$(a\psi_0 + b\psi_1)^\dagger(c\psi_0 + d\psi_1) = a^\dagger c + b^\dagger d.$$

Theorem [8.36]
(eigenMinus_goodBad_isNormalized)
Let $n \in \mathbb{N}$ *and* ψ_0, ψ_1 *be orthonormal vectors of dimension* 2^n*. Then, the vector* $\frac{1}{\sqrt{2}}(\psi_0 - i\psi_1)$ *is normalized.*

Theorem [8.37]
(eigenPlus_goodBad_isNormalized)
Let $n \in \mathbb{N}$ *and* ψ_0, ψ_1 *be orthonormal vectors of dimension* 2^n*. Then, the vector* $\frac{1}{\sqrt{2}}(\psi_0 + i\psi_1)$ *is normalized.*

Theorem [8.38]
(eigenMinus_inner_eigenPlus_goodBad)
Let $n \in \mathbb{N}$ and ψ_0, ψ_1 be orthonormal vectors of dimension 2^n. Vectors $\frac{1}{\sqrt{2}}(\psi_0 - i\psi_1)$ and $\frac{1}{\sqrt{2}}(\psi_0 + i\psi_1)$ are orthogonal.

Theorem [8.39]
(paperQAEOutputProbability_eq_good_bad_mixture)
Let $m, n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n, $f : \mathbb{Z}_{2^n} \to \{0, 1\}$, ψ_0, ψ_1 vectors of dimension 2^n, $\theta \in \mathbb{R}$, and $y \in \mathbb{Z}_{2^m}$. Assume that

$$A |0\rangle = \cos(\theta)\psi_0 + \sin(\theta)\psi_1.$$

Assume also that $\psi_{0k} = 0$ for all k such that $f(k) = 1$, while $\psi_{1k} = 0$ for all k such that $f(k) = 0$. Moreover, assume that ψ_0 and ψ_1 are normalized.

Then,

$$\langle y| E_4(m, n, A, f) |y\rangle = \frac{1}{2}||a(m, \frac{\theta}{\pi}, y)||^2 + \frac{1}{2}||a(m, 1 - \frac{\theta}{\pi}, y)||^2.$$

Definition [8.22]
(qaeGroverPlaneMarginal)
Let $m \in \mathbb{N}$, $\theta \in \mathbb{R}$, $y \in \mathbb{Z}_{2^m}$, and

$$\psi = (IQ(m) \otimes I_2) \times \Lambda(R_y(4\theta), m) \times (|u(m)\rangle \otimes (\cos(\theta) |0\rangle + \sin(\theta) |1\rangle)).$$

$$PM(m, y, \theta) = \sum_{j=0}^{1} ||\psi_{j+2y}||^2.$$

Theorem [8.40]
(qaeGroverPlaneMarginal_eq)
Let $m \in \mathbb{N}$, $\theta \in \mathbb{R}$, $y \in \mathbb{Z}_{2^m}$, and

$$\psi = (IQ(m) \otimes I_2) \times \Lambda(R_y(4\theta), m) \times (|u(m)\rangle \otimes (\cos(\theta) |0\rangle + \sin(\theta) |1\rangle)).$$

Then,

$$\sum_{j=0}^{1} ||\psi_{j+2y}||^2 = \frac{1}{2}||a(m, \frac{\theta}{\pi}, y)||^2 + \frac{1}{2}||a(m, -\frac{\theta}{\pi}, y)||^2.$$

Theorem [8.41]
(amplitudeFromAngle_pi_sub)
Let $\theta \in \mathbb{R}$. Then,

$$\sin^2(\pi - \theta) = \sin^2(\theta).$$

Theorem [8.42]
(qaeGroverPlaneMarginal_eq_wrapped)
Let $m \in \mathbb{N}$, $\theta \in \mathbb{R}$, $y \in \mathbb{Z}_{2^m}$, and

$$\psi = (IQ(m) \otimes I_2) \times \Lambda(R_y(4\theta), m) \times (|u(m)\rangle \otimes (\cos(\theta)\,|0\rangle + \sin(\theta)\,|1\rangle)).$$

Then,

$$\sum_{j=0}^{1} ||\psi_{j+2y}||^2 = \frac{1}{2}||a(m, \frac{\theta}{\pi}, y)||^2 + \frac{1}{2}||a(m, 1 - \frac{\theta}{\pi}, y)||^2.$$

Theorem [8.43]
(phase_angle_error_of_pos_window)
Let $m, k \in \mathbb{N}$, $\theta \in \mathbb{R}$, and $y \in \mathbb{Z}_{2^m}$. Assume that

$$\left|\frac{\theta}{\pi} - \frac{y}{2^m}\right| \leq \frac{k}{2^m}.$$

Then,

$$\left|\theta - \pi\frac{y}{2^m}\right| \leq \pi\frac{k}{2^m}.$$

Definition [8.23]
(qaeGroverPlaneSuccessfulOutcomesK)
Let $m, k \in \mathbb{N}$, $\theta \in \mathbb{R}$, $a = \sin^2(\theta)$, *and* $M = 2^m$.

$$S(m,k,\theta) = \left\{ y \in \mathbb{Z}_{2^m} \,\middle|\, \left| \sin^2\left(\pi \frac{y}{M}\right) - a \right| \leq 2\pi \frac{k}{M} \sqrt{a(1-a)} + \pi^2 \frac{k^2}{M^2} \right\}$$

Definition [8.24]
(qaeGroverPlaneSuccessProbabilityK)
Let $m, k \in \mathbb{N}$ *and* $\theta \in \mathbb{R}$.

$$P(m,k,\theta) = \sum_{y \in S(m,k,\theta)} PM(m,y,\theta).$$

Theorem [8.44]
(mem_qaeGroverPlaneSuccessfulOutcomesK_of_estimate_error)
Let $m, k \in \mathbb{N}$, $\theta \in \mathbb{R}$, $y \in \mathbb{Z}_{2^m}$, $a = \sin^2(\theta)$, *and* $M = 2^m$. *Then,*

$$y \in S(m,k,\theta) \iff \left| \sin^2\left(\pi \frac{y}{M}\right) - a \right| \leq 2\pi \frac{k}{M} \sqrt{a(1-a)} + \pi^2 \frac{k^2}{M^2}.$$

Theorem [8.45]
(amplitudeFromAngle_sub_pi)
Let $\theta \in \mathbb{R}$. *Then,*

$$\sin^2(\theta - \pi) = \sin^2(\theta).$$

Theorem [8.46]
(amplitudeFromAngle_add_pi)
Let $\theta \in \mathbb{R}$. *Then,*

$$\sin^2(\theta + \pi) = \sin^2(\theta).$$

Theorem [8.47]
(phase_angle_error_of_window_sub_one)
Let $m, k \in \mathbb{N}$, $\alpha \in \mathbb{R}$, *and* $y \in \mathbb{Z}_{2^m}$. *Assume that*

$$\left|\frac{\alpha}{\pi} - \frac{y}{2^m} - 1\right| \le \frac{k}{2^m}.$$

Then,

$$\left|\pi\frac{y}{2^m} - (\alpha - \pi)\right| \le \pi\frac{k}{2^m}.$$

Theorem [8.48]
(phase_angle_error_of_window_add_one)
Let $m, k \in \mathbb{N}$, $\alpha \in \mathbb{R}$, and $y \in \mathbb{Z}_{2^m}$. Assume that

$$\left|\frac{\alpha}{\pi} - \frac{y}{2^m} + 1\right| \le \frac{k}{2^m}.$$

Then,

$$\left|\pi\frac{y}{2^m} - (\alpha + \pi)\right| \le \pi\frac{k}{2^m}.$$

Theorem [8.49]
(estAmp_error_of_phase_circular_window)
Let $m, k \in \mathbb{N}$, $\alpha \in \mathbb{R}$, $y \in \mathbb{Z}_{2^m}$, $a = \sin^2(\alpha)$, and $M = 2^m$. Assume that

$$d\left(\frac{\alpha}{\pi}, \frac{y}{2^m}\right) \le \frac{k}{2^m}.$$

Then,

$$\left|\sin^2\left(\pi\frac{y}{2^m}\right) - a\right| \le 2\pi\frac{k}{M}\sqrt{a(1-a)} + \pi^2\frac{k^2}{M^2}.$$

Theorem [8.50]
(qaeGroverPlane_estimate_error_of_wrapped_neg_circular_window)
Let $m, k \in \mathbb{N}$, $\theta \in \mathbb{R}$, $y \in \mathbb{Z}_{2^m}$, $a = \sin^2(\theta)$, and $M = 2^m$. Assume that

$$d\left(1 - \frac{\theta}{\pi}, \frac{y}{2^m}\right) \le \frac{k}{2^m}.$$

Then,

$$\left|\sin^2\left(\pi\frac{y}{2^m}\right) - a\right| \leq 2\pi\frac{k}{M}\sqrt{a(1-a)} + \pi^2\frac{k^2}{M^2}.$$

Theorem [8.51]
(qaeGroverPlane_pos_circular_window_subset_success)
Let $m, k \in \mathbb{N}$ and $\theta \in \mathbb{R}$. Then,

$$O_{m,k,\frac{\theta}{\pi}} \subseteq S_{m,k,\theta}.$$

Corollary
(qaeGroverPlane_wrapped_neg_circular_window_subset_success)
The following holds.

$$O_{m,k,1-\frac{\theta}{\pi}} \subseteq S_{m,k,\theta}.$$

Theorem [8.52]
(qaeGroverPlaneSuccessProbabilityK_lower_bound_of_qpe_circular_windows)
Let $m, k \in \mathbb{N}$ and $\theta \in \mathbb{R}$. Assume that the following two propositions hold.

$$\begin{cases} \frac{8}{\pi^2}, & k = 1 \\ 1 - \frac{1}{2(k-1)}, & otherwise \end{cases} \leq Pr\{y \in O_{m,k,\frac{\theta}{\pi}} | Y\},$$
$$\begin{cases} \frac{8}{\pi^2}, & k = 1 \\ 1 - \frac{1}{2(k-1)}, & otherwise \end{cases} \leq Pr\{y \in O_{m,k,1-\frac{\theta}{\pi}} | Y\}.$$

Then,

$$\begin{cases} \frac{8}{\pi^2}, & k = 1 \\ 1 - \frac{1}{2(k-1)}, & otherwise \end{cases} \leq P(m,k,\theta).$$

Theorem [8.53]
(qaeGroverPlaneSuccessProbabilityK_lower_bound)
Let $m, k \in \mathbb{N}$ and $\theta \in [0, \pi]$. Assume that $k > 0$. Then,

$$\begin{cases} \frac{8}{\pi^2}, & k = 1 \\ 1 - \frac{1}{2(k-1)}, & otherwise \end{cases} \leq P(m,k,\theta).$$

Theorem [8.54]
(theorem12ErrorBound_eq_expanded_QPE)
Just a simple unfolding of the definition of theorem12ErrorBound.

Theorem [8.55]
(paperQAESuccessProbabilityK_lower_bound)
Let $m, n, k \in \mathbb{N}$, A be a unitary matrix of dimension 2^n, $f : \mathbb{Z}_{2^n} \to \{0, 1\}$, ψ_0, ψ_1 vectors of dimension 2^n, and $\theta \in [0, \pi]$. Assume that $k > 0$, ψ_0 and ψ_1 are normalized, that ${\psi_0}_k = 0$ for all k such that $f(k) = 1$, and ${\psi_1}_k = 0$ for all k such that $f(k) = 0$. Assume also that

$$A\left|0\right\rangle = \cos(\theta)\psi_0 + \sin(\theta)\psi_1.$$

Then,

$$\begin{cases} \frac{8}{\pi^2}, & k = 1 \\ 1 - \frac{1}{2(k-1)}, & \textit{otherwise} \end{cases} \leq \sum_{y \in S(m,k,\theta)} \left\langle y\right| E_4(m, n, A, f) \left|y\right\rangle.$$

Corollary
(paperQAESuccessProbabilityK_lower_bound_expanded)
The same as the theorem above, except that $S(m, k, \theta)$ is unfolded.

Theorem [8.56]
(qpeApproxOutcomeProbability_exact_grid_eq_one)
Let $m \in \mathbb{N}$ and $y \in \mathbb{Z}_{2^m}$. Then,

$$||a(m, \frac{y}{2^m}, y)||^2 = 1.$$

Theorem [8.57]
(paperQAEHalfGridRatio)
Let $m \in \mathbb{N}$ such that $m > 0$. Then,

$$\frac{2^{m-1}}{2^m} = \frac{1}{2}.$$

Theorem [8.58]
(paperGroverOperator_mul_prepared_of_goodProbability_zero)

Let $n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n, and $f : \mathbb{Z}_{2^n} \to \{0,1\}$. Assume that

$$\sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \langle k| A |0\rangle \langle 0| A^\dagger |k\rangle = 0.$$

Then,

$$G_f(A) \times A |0\rangle = A |0\rangle .$$

Theorem [8.59] (paperGroverOperator_mul_prepared_of_badProbability_zero)
Let $n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n, and $f : \mathbb{Z}_{2^n} \to \{0,1\}$. Assume that

$$\sum_{\substack{k=0 \\ f(k)=0}}^{2^n-1} \langle k| A |0\rangle \langle 0| A^\dagger |k\rangle = 0.$$

Then,

$$G_f(A) \times A |0\rangle = e^{\pi i} A |0\rangle .$$

Theorem [8.60] (paperQAEEndpointZero_of_goodProbability_zero)
Let $n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n, and $f : \mathbb{Z}_{2^n} \to \{0,1\}$. Assume that

$$\sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \langle k| A |0\rangle \langle 0| A^\dagger |k\rangle = 0.$$

Then, there exists y such that

$$\langle y| E_4(m, n, A, f) |y\rangle = 1.$$

Moreover, for such y,

$$\sin^2\left(\pi\frac{y}{2^m}\right) = 0.$$

Theorem [8.61]
(paperQAEEndpointOne_of_badProbability_zero)
Let $m, n \in \mathbb{N}$, A be a unitary matrix of dimension 2^n, and $f : \mathbb{Z}_{2^n} \to \{0, 1\}$. Assume that $m > 0$ and that

$$\sum_{\substack{k=0 \\ f(k)=0}}^{2^n-1} \langle k| A |0\rangle \langle 0| A^\dagger |k\rangle = 0.$$

Then, there exists y such that

$$\langle y| E_4(m, n, A, f) |y\rangle = 1.$$

Moreover, for such y,

$$\sin^2\left(\pi\frac{y}{2^m}\right) = 1.$$

Theorem [8.62]
(paperQAEOutputProbability_nonneg)
Let $m, n \in \mathbb{N}$, A be a square matrix of dimension 2^n, $f : \mathbb{Z}_{2^n} \to \{0, 1\}$, and $y \in \mathbb{Z}_{2^m}$. Then,

$$\langle y| E_4(m, n, A, f) |y\rangle \geq 0.$$

Theorem [8.63]
(paperQAESuccessSum_ge_one_of_point)
Let $m, n, k \in \mathbb{N}$, A be a square matrix of dimension 2^n, $f : \mathbb{Z}_{2^n} \to \{0, 1\}$, $a \in \mathbb{R}$, $M = 2^m$, and $y \in \mathbb{Z}_{2^m}$. Assume that

$$\langle y| E_4(m, n, A, f) |y\rangle = 1.$$

Moreover, assume that

$$|\sin^2\left(\pi\frac{y}{2^m}\right) - a| \leq 2\pi\frac{k}{M}\sqrt{a(1-a)} + \pi^2\frac{k^2}{M^2}.$$

Then,

$$1 \leq \sum_{y\in S(m,k,\theta)} \langle y| E_4(m,n,A,f) |y\rangle .$$

Theorem [8.64]
(theorem12SuccessProbability_le_one)
Let $k \in \mathbb{N}$ such that $k > 0$, then

$$\begin{cases} \frac{8}{\pi^2}, & k = 1 \\ 1 - \frac{1}{2(k-1)}, & \textit{otherwise} \end{cases} \leq 1.$$

Theorem [8.65]
(PaperTheorem12)
Let $m, n, k \in \mathbb{N}$, A be a unitary matrix of dimension 2^n, $f : \mathbb{Z}_{2^n} \to \{0,1\}$, $\theta \in [0,\pi]$, $a = \sin^2(\theta)$, $m > 0$, and $k > 0$. Assume that

$$\sum_{\substack{k=0 \\ f(k)=1}}^{2^n-1} \langle k| A |0\rangle \langle 0| A^\dagger |k\rangle = a.$$

Then, the following four propositions hold.

1. *If $k = 1$, then*

$$\frac{8}{\pi^2} \leq \sum_{y\in S(m,k,\theta)} \langle y| E_4(m,n,A,f) |y\rangle .$$

2. *If $k > 1$, then*

$$1 - \frac{1}{2(k-1)} \leq \sum_{y\in S(m,k,\theta)} \langle y| E_4(m,n,A,f) |y\rangle .$$

3. If $a = 0$, then there exists $y \in \mathbb{Z}_{2^m}$ such that

$$\langle y| E_4(m, n, A, f) |y\rangle = 1.$$

Moreover, for such y, $\sin^2(\pi \frac{y}{2^m}) = 0$.

4. If $a = 1$, then there exists $y \in \mathbb{Z}_{2^m}$ such that

$$\langle y| E_4(m, n, A, f) |y\rangle = 1.$$

Moreover, for such y, $\sin^2(\pi \frac{y}{2^m}) = 1$.